Vehicle Studio Image Processing

Technical overview for integration teams | Version 1.0 | 19 October 2026

1. Purpose

This document describes the service that turns customer vehicle photos into clean, studio-style images on a transparent background, ready for your 360° / 3D viewer and listing pipeline. It covers what the service does to each image, the available service tiers, input and output specifications, turnaround and known limitations. The API specification (endpoints, authentication and payloads) is provided separately.

2. Processing overview

Every image passes through a proprietary, multi-stage AI processing pipeline:

Stage	What happens
1. Intake & normalisation	Format conversion (JPG, PNG, WEBP, HEIC), orientation correction from camera metadata, size normalisation.
2. Vehicle isolation	The vehicle is separated from its surroundings. Background, ground, people and the vehicle's ground shadow are removed; the result is placed on a fully transparent background.
3. Glass treatment	Window glass receives dedicated treatment. Reflections of sky, trees and buildings, and any scenery visible through the vehicle, are removed. Glass is rendered as uniform, light-grey tinted glass so every image of a vehicle looks consistent.
4. Lighting	Depending on the service tier, either the photo's original lighting is kept, or the vehicle is relit with neutral, even studio lighting (no sunset tint, harsh sun or hard shadows).
5. Condition preservation	The vehicle's real condition is kept: scratches, dents, scuffs, dirt, wear, stickers, badges and licence plates remain visible. Nothing is cleaned or repaired.
6. Delivery & status	Each image is returned as a PNG with a per-image status. Temporary processing errors are retried automatically; an image that still fails is reported individually and the rest of the vehicle set completes.

3. Service tiers

Tier	Lighting	Typical use
Standard	Original lighting of the photo is kept	High volume listings; fastest and most economical
Studio	Neutral, even studio lighting	Premium listings and 3D/360° viewers where a uniform studio look across all angles matters
Studio Plus	As Studio, rendered at higher detail	Hero images and marketing material

All tiers include vehicle isolation, transparent background and glass treatment.

4. Supported photo types

- **Exterior**: all angles of the vehicle (front, rear, sides, three-quarter views). A typical set is 12 photos per vehicle.
- **Interior / open door**: cabin views, dashboard, seats and open-door shots. The cabin is kept as photographed; everything outside the vehicle (the view through the windscreen, windows, sunroof and open doors) is removed.
- The photo type is specified per image when submitting a job.

5. Input requirements

Item	Requirement
Formats	JPG, PNG, WEBP; HEIC (iPhone) supported
Resolution	Minimum 1024 px on the long side; standard phone photos (12 MP+) recommended
Framing	The whole vehicle in frame with some margin; consistent distance and height across the set
Lighting	Daylight or even artificial light; avoid strong backlight directly behind the vehicle
Content	One vehicle per photo; avoid people leaning on or standing in front of the vehicle
Rights	Only photos you own or are licensed to process. No watermarked or third-party stock images

6. Output specification

Item	Specification
Format	PNG, RGBA (8-bit per channel) with a transparent background
Resolution	Approximately 1.5 megapixels per image (landscape about 1536 × 1024 px), independent of input size
Glass	Uniform light-grey tinted glass, optimised for white or light studio backgrounds. Glass is rendered solid (not see-through)
Naming	Output files keep the input file name with a .png extension, grouped per vehicle
Per-image metadata	Status (ok / failed), processing time, tier and photo type

7. Turnaround

- About 20–25 seconds per image.
- Images of a vehicle are processed in parallel: a 12-photo set typically completes in about 1.5–2 minutes.
- Processing is asynchronous: a job is submitted, then results are retrieved when ready (status polling or callback, as defined in the API specification).

8. Infrastructure

- Every image is processed on **GPU-accelerated cloud AI infrastructure**. Each image uses dedicated GPU compute for vehicle isolation, glass treatment and relighting; this compute is the main per-image cost driver.
- Capacity scales with demand, so turnaround stays consistent for large batches and peak periods.
- Higher tiers use more GPU compute per image (higher rendering detail), which is reflected in tier pricing.

9. Example results



Original photo



Standard tier: original lighting, transparent background, treated glass



Studio tier: neutral studio lighting, same condition preserved

Output shown on a white background. Delivered files have a transparent background.

10. Known limitations

- **AI-rendered output.** Images are re-rendered by AI, not retouched pixel by pixel. Fine details (wheel spokes, small badges, panel gaps) are reproduced closely but may differ slightly between angles of the same vehicle. Please validate with your 3D/360° stitching on a sample set.
- **Condition.** Damage and wear are preserved and clearly visible, but not as a pixel-exact copy. Keep the original photos where exact damage documentation is required (for example, inspections or claims).
- **Studio lighting and colour.** Removing strong coloured light (for example, golden-hour sun) can make materials appear in a more neutral tone than in the original photo.
- **Resolution.** Output is about 1.5 MP regardless of the input resolution.
- **Rare mismatches.** In rare cases an image may not match the input vehicle exactly. Such images can be re-processed on request.

11. Data handling

- Images are transferred over encrypted connections (HTTPS) and processed on secure cloud AI infrastructure, including third-party AI processing providers.
- Images are used only to produce your results. Retention and deletion periods are defined in the service agreement.
- Photos may contain licence plates and people. Please make sure your customers' consent and your privacy notice cover image processing.

For integration questions, please contact your account manager.